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PROJECT REPORT

SonarShield

Smart Sonar Radar & Safety Alert System

An Arduino-based smart surveillance prototype featuring servo-controlled sonar radar scanning, real-time object distance detection, ambient light sensing, automatic and manual light control, proximity-based LED and buzzer alerts, LCD status display, and interactive PC radar visualization.

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Submitted to:

Sameer Khairul

Lecturer, Dept. of Information and Communication Engineering
Daffodil International University

Submitted by:

Nazmul Ahmed Fahim (242-50-040)

Mst. Amena Khatun (242-50-048)

Fatin Ishrak (242-50-053)

Tasnim Imam Fema (242-50-054)

Jannatul Mawoa (242-50-055)

Md. Abdulla Hasan (242-50-058)

Zaffar Abdullah (242-50-059)

Md. Junayed Ahmed (242-50-060)

Dept. of Information and Communication Engineering
Daffodil International University

1. Project Title

SonarShield: Smart Sonar Radar & Safety Alert System is an Arduino Uno-based embedded prototype that combines a servo-driven ultrasonic radar, ambient light sensing, and a live Processing-based dashboard into one platform. It continuously scans its surroundings, measures object distance, reports light conditions, and raises visual/audible alerts when an object comes too close with key controls (buzzer mute, light mode) operable directly from the desktop GUI.

2. Purpose

Objective

- To build a rotating ultrasonic radar using an HC-SR04 sensor and an SG90 servo motor.
- To generate a proximity-based safety alert using a Red LED (visual) and a buzzer (audible).
- To sense ambient brightness with an LDR and automatically control a Light LED, with a manual override from the GUI.
- To build a real-time Processing dashboard that visualizes the radar sweep and sends control commands back to the Arduino.
- To display live status locally on a 16×2 I2C LCD, independent of the computer.
- To gain hands-on experience with sensor interfacing, PWM servo control, I2C communication, and bidirectional serial protocol design.

Scope

The current prototype targets indoor, short-range demonstration (reliable object detection up to about 40 cm) over a wired USB-serial link. It covers ultrasonic distance sensing, servo-driven sweep, ambient light sensing, local LCD feedback, and PC-based radar visualization with two-way control. Outdoor use, long-range detection, wireless connectivity, and industrial ruggedization are outside the current scope and are noted as future directions.

Significance

SonarShield demonstrates how several peripherals — an ultrasonic sensor, a servo, an LDR, LEDs, a buzzer, and an LCD can be coordinated by a single microcontroller while staying synchronized with an external monitoring application. It gives a practical, low-cost foundation for the sensor-fusion and two-way hardware-software communication skills used in more advanced robotics and monitoring systems.

3. Components

SonarShield is built entirely from low-cost, easily available components. Table 1 lists every part used and its role in the system.

Component	Qty	Purpose
Arduino Uno R3	1	Main microcontroller; runs all sensing and control logic
HC-SR04 Ultrasonic Sensor	1	Measures object distance (radar sensing)
SG90 Servo Motor	1	Rotates the ultrasonic sensor for the radar sweep
16×2 I2C LCD Display	1	Shows live distance, light, and buzzer status locally

Component	Qty	Purpose
LDR Light Sensor Module	1	Detects ambient brightness / darkness
White LED Module	1	Auto-ON in the dark, or manually forced via the GUI
Red LED Module	1	Steady ON whenever an object is within 40 cm
Active Buzzer Module	1	Sounds only when an object is within 15 cm; mutable from the GUI
Breadboard	1	Circuit assembly
Jumper Wires	Several	Circuit interconnection
USB Cable	1	Power and serial communication with the PC
Cardboard & Glue Stick	1 set	Physical mounting for the radar assembly

Key operating specifications: HC-SR04 effective range $\approx 2\text{--}400\text{ cm}$; servo sweep $15^\circ\text{--}190^\circ$; LCD I2C address $0x27$; USB-serial link at 9600 baud.

4. Diagram / Circuit Setup

The circuit was designed and verified in Cirkuit Designer before assembly. The Arduino Uno acts as the central hub: the HC-SR04 and LDR feed sensor data in, while the servo, LEDs, buzzer, and LCD receive control signals out. The LCD communicates over the I2C bus, keeping only two pins occupied for a full alphanumeric display.

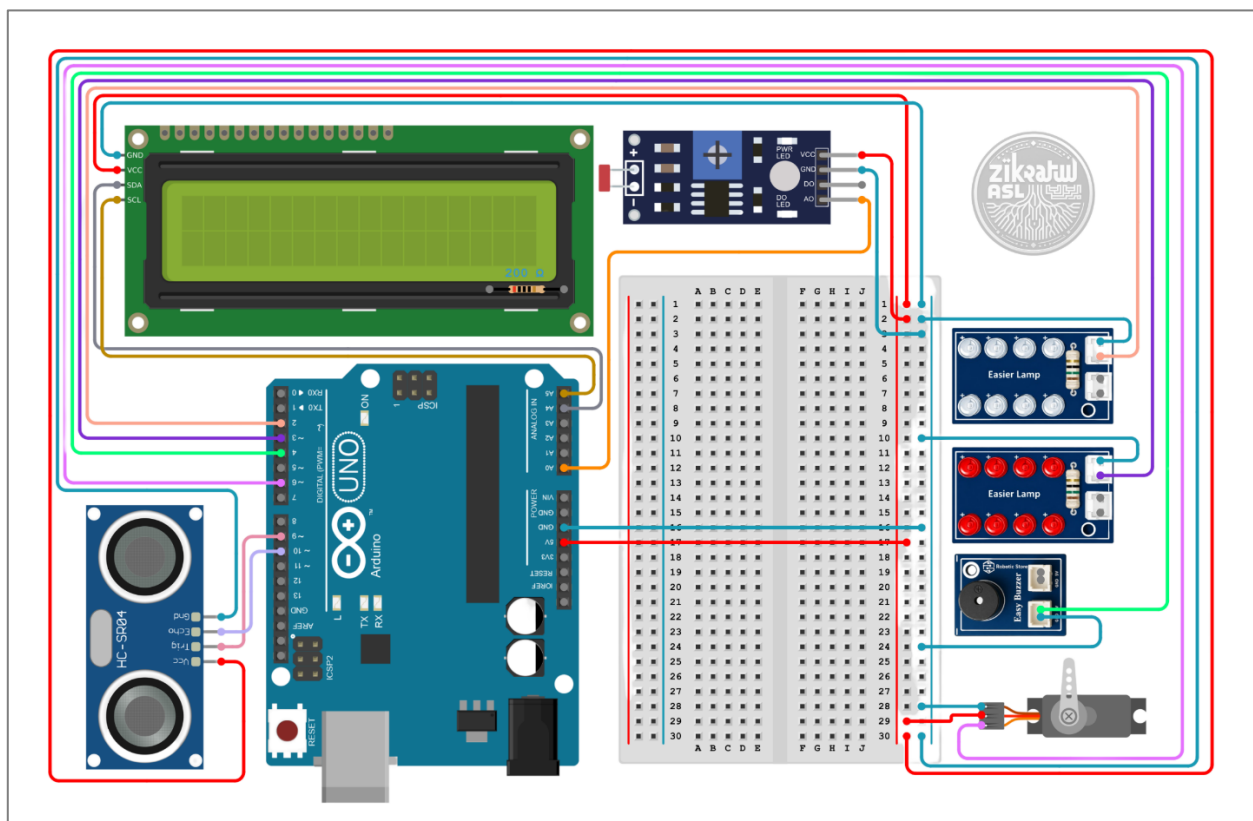


Figure 1: SonarShield circuit diagram (Cirkuit Designer)

Pin Connection Summary

Peripheral	Arduino Pin(s)	Notes
HC-SR04 Trig / Echo	D9 / D10	5V & GND shared from Arduino power rails
Servo signal	D6	PWM-capable pin required
Light LED (+220 Ω)	D2	Auto ON in dark, or forced via GUI
Red LED (+220 Ω)	D3	ON whenever object is within 40 cm
Buzzer	D4	ON only within 15 cm; mutable from GUI
LDR	A0	Voltage-divider circuit with a fixed resistor to GND
LCD (I2C)	A4 (SDA), A5 (SCL)	Address 0x27

A USB cable connects the Arduino to the host PC, carrying the serial data stream that drives the Processing GUI and, in the reverse direction, the mute/light-mode commands sent back from that GUI making the link fully bidirectional. Figure 2 summarizes how data moves through the whole system.

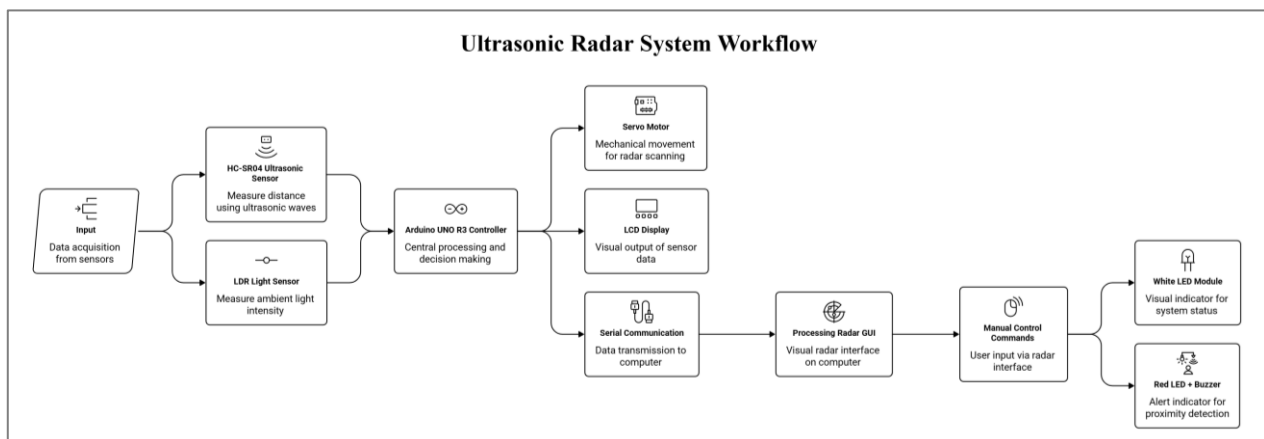


Figure 2: SonarShield system workflow

5. Cost Breakdown

Component	Price (BDT)
Arduino Uno	850/-
Breadboard (Medium)	100/-
LCD Display (I2C 16×2)	300/-
Ultrasonic Sensor	125/-
Servo Motor (SG90)	125/-
Buzzer Module	50/-
Glue Stick	20/-
Jumper Wires & Others	100/-
LED Module (Red + White)	100/-
LDR Sensor Module	80/-
Cardboard	50/-

Total Estimated Cost: 1,900 BDT. All major parts (Arduino, sensors, LCD, servo) are reusable in future projects, so the effective cost is even lower.

6. Functionality Breakdown

Functionality 1: Ultrasonic Radar Scanning & Proximity Alert

The servo continuously sweeps the HC-SR04 across a 15°–190° arc. At every step, distance is measured and streamed to the PC, where the Processing dashboard renders a live radar sweep with a fading afterglow trail and a red ‘detection wedge’ whenever an object falls within 40 cm. The Red LED mirrors this as a steady hardware indicator, while the buzzer activates only within 15 cm — and can be muted directly from the GUI without touching the hardware.

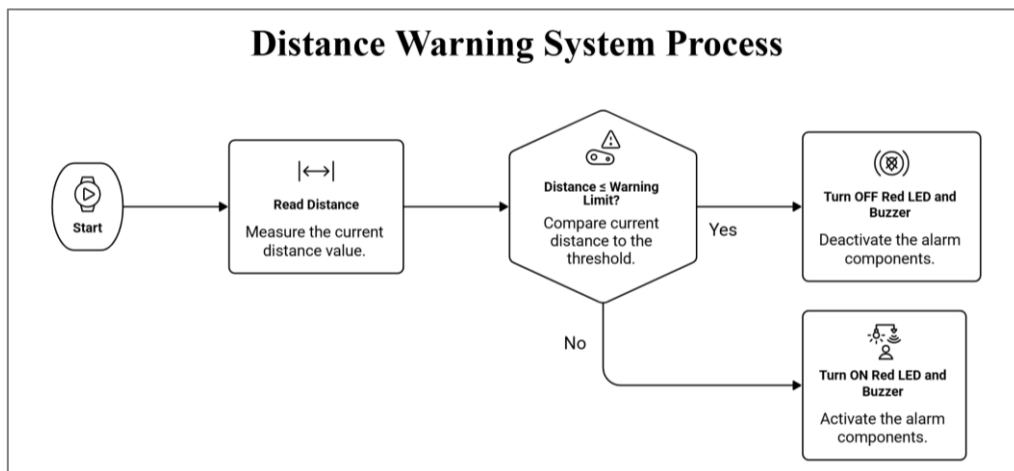


Figure 3: Proximity alert decision logic

Functionality 2: Ambient Light Sensing & Adaptive Lighting

An LDR continuously monitors ambient brightness. In AUTO mode the Light LED switches on automatically when it gets dark and off in bright conditions. A GUI button cycles the light through AUTO, FORCED ON, and FORCED OFF, sending the corresponding command back to the Arduino over serial.

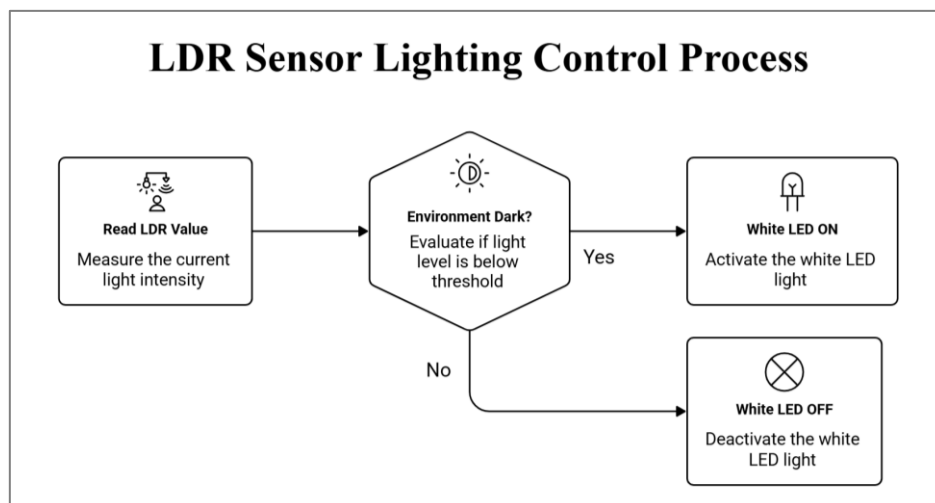


Figure 4: Adaptive lighting decision logic

Functionality 3: Real-Time GUI with Bidirectional Control

The Processing application parses the Arduino's serial stream and renders the radar sweep, angle, distance, light status, and buzzer status in real time. It is bidirectional: button presses in the GUI send single-character commands ('M' to mute/unmute the buzzer, 'L' to cycle the light mode) back to the Arduino, so the hardware can be controlled without touching it directly.

Functionality 4: On-Device LCD Status Display

The 16×2 I2C LCD mirrors the current distance and light/buzzer status locally, so the system remains readable even when the PC dashboard is not being watched.

7. Business Proposal

As an initial-stage prototype, SonarShield is positioned as a low-cost ($\approx$ 1,900 BDT) educational and light-security demonstrator. In its current form it suits small-scale use cases a desktop proximity alarm, a doorway/parking obstacle sensor, or a hands-on STEM kit for microprocessor and sensor-interfacing courses where an affordable radar-style sensor with live visualization adds clear value over a single-sensor alarm. Likely early adopters include students and hobbyists, small offices/labs wanting a low-cost entry monitor, and warehouses needing a simple obstacle-awareness add-on.

Looking beyond this version, the project is meant as a foundation rather than a finished product. Its core techniques servo-actuated sensor sweeping, sensor fusion, and closed-loop communication between hardware and a monitoring dashboard are the same building blocks used in more advanced platforms. With further development (higher-precision ranging, wireless connectivity, a mobile/cloud dashboard, and sturdier mechanical design), the architecture could extend toward autonomous obstacle-avoidance robotics and, longer-term, toward research-oriented perimeter surveillance or defense-adjacent training systems where multi-sensor tracking and automated response are core requirements. This roadmap is presented at a conceptual, business-planning level only; realizing it would require substantial further engineering, safety validation, and domain expertise beyond this lab project.

8. Potential Challenges

- The HC-SR04's range and accuracy are limited compared with industrial radar/LiDAR, and soft or angled surfaces can cause missed or false echoes.
- The LDR's dark/bright threshold is location-specific and needs recalibration under different lighting.
- Continuous servo sweeping over long periods can mechanically wear the motor faster than industrial-grade scanning mechanisms.
- The system depends on a wired USB-serial link, so it must stay physically connected to a PC; the servo's current draw can also briefly disturb other peripherals when powered solely from USB.
- The cardboard-mounted prototype is suitable for lab demonstration only — a real product would need a durable enclosure and proper PCB design.
- Scaling toward outdoor, industrial, or security/defense-grade use would require environmental ruggedization plus formal safety and regulatory validation well beyond an academic prototype.